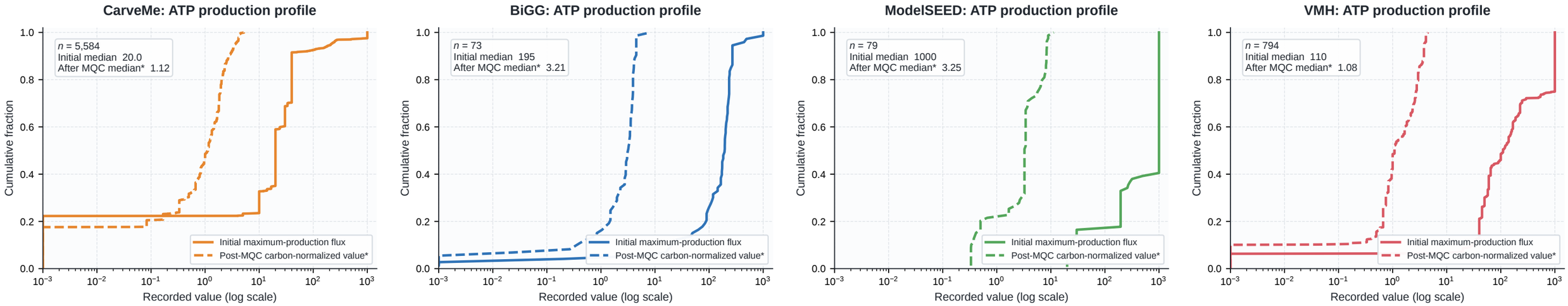
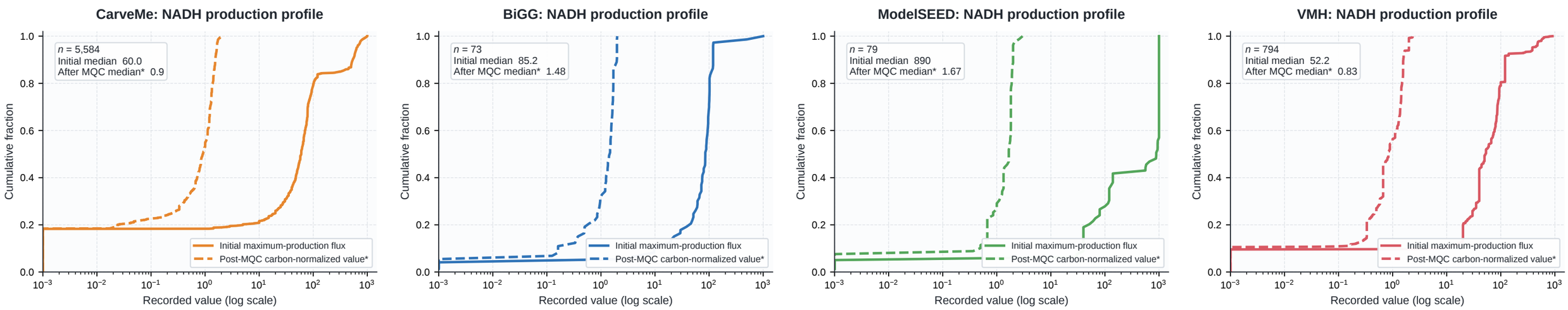
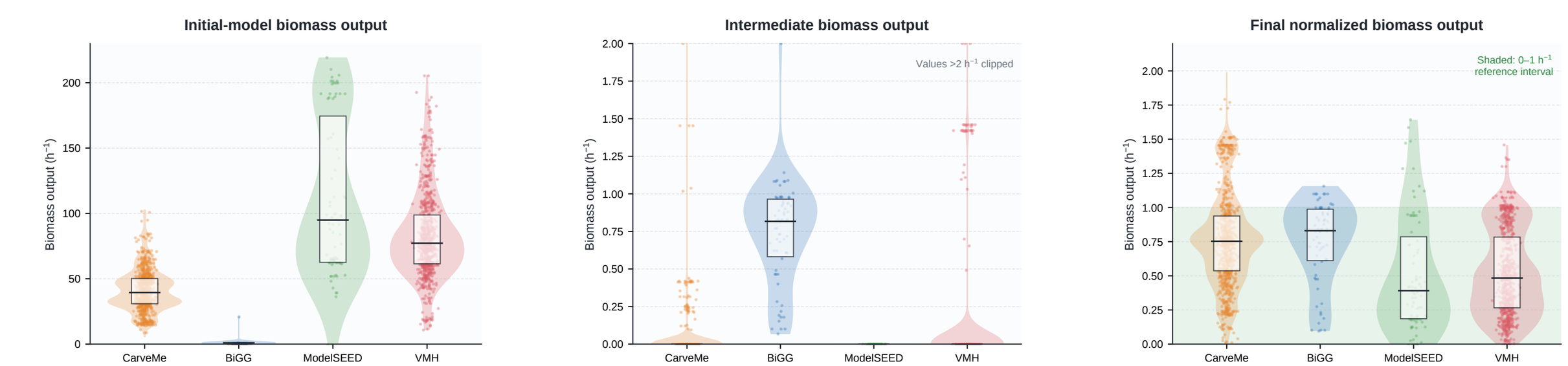
a


CarveMe: ATP production profile
n = 5,584
Initial median 20.0
After MQC median* 1.12
BiGG: ATP production profile
n = 73
Initial median 195
After MQC median* 3.21
ModelSEED: ATP production profile
n = 79
Initial median 1000
After MQC median* 3.25
VMH: ATP production profile
n = 794
Initial median 110
After MQC median* 1.08
Cumulative fraction
Recorded value (log scale)
Initial maximum-production flux
Post-MQC carbon-normalized value*

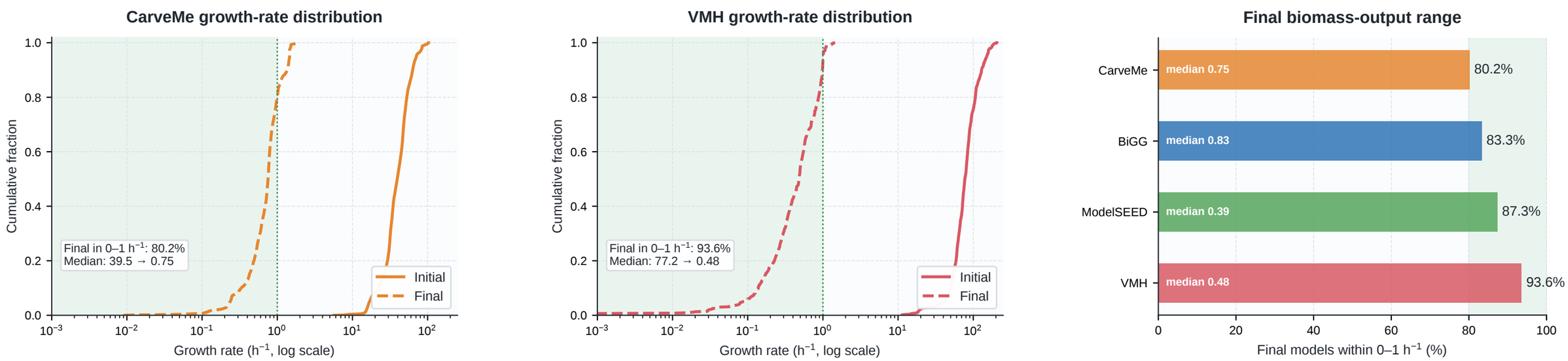
b


CarveMe: NADH production profile
n = 5,584
Initial median 60.0
After MQC median* 0.9
BiGG: NADH production profile
n = 73
Initial median 85.2
After MQC median* 1.48
ModelSEED: NADH production profile
n = 79
Initial median 890
After MQC median* 1.67
VMH: NADH production profile
n = 794
Initial median 52.2
After MQC median* 0.83
Cumulative fraction
Recorded value (log scale)
Initial maximum-production flux
Post-MQC carbon-normalized value*

c


Initial-model biomass output
Intermediate biomass output
Values >2 h⁻¹ clipped
Final normalized biomass output
Shaded: 0–1 h⁻¹ reference interval
Biomass output (h⁻¹)
CarveMe
BiGG
ModelSEED
VMH

d


CarveMe growth-rate distribution
Final in 0–1 h⁻¹: 80.2%
Median: 39.5 → 0.75
VMH growth-rate distribution
Final in 0–1 h⁻¹: 93.6%
Median: 77.2 → 0.48
Initial
Final
Cumulative fraction
Growth rate (h⁻¹, log scale)
Final biomass-output range
median 0.75
80.2%
median 0.83
83.3%
median 0.39
87.3%
median 0.48
93.6%
Final models within 0–1 h⁻¹ (%)